

BuildShip Intern Task: Smart Lead Intake API

A hands-on, 3 to 4 day project to build and ship your first AI-powered backend workflow.

Task at a glance

Read this page first. It tells you what you are building, how long it should take and what you need.

Task name	Smart Lead Intake API
Assigned to	Intern (Automation / Backend track)
Difficulty	Medium. No prior BuildShip experience needed, but you should be comfortable reading JSON and writing a few lines of JavaScript.
Duration	3 to 4 working days
Cost	Free. Everything runs on a free BuildShip account, a free Google account and a free-tier AI API key.
Main tool	BuildShip (visual backend and workflow builder)
Reports to	Your Devowise Studios mentor

The scenario

Visitors fill in a contact form on a client website. Today, someone has to read every message by hand, decide how urgent it is and copy the details into a spreadsheet. This is slow, and hot leads can sit unanswered for hours.

Your job is to build a small backend API in BuildShip that does this automatically. When a form is submitted, your API checks the data, asks an AI model to classify the lead, saves it to a Google Sheet and alerts the team if the lead looks hot. It then sends a clean JSON answer back to the website.

Why this task? It touches the five things every automation project needs: an API trigger, input validation, an AI step, saving data and sending notifications. Finish it and you can build most workflows we deliver to clients.

Goals, scope and setup

What success looks like, what is in and out of scope, and what to prepare before you start.

Learning goals

- Create a REST API endpoint in BuildShip and test it with Postman or curl.
- Validate and clean incoming data with a Script node.
- Call an AI model and force it to return structured JSON.
- Connect to Google Sheets and send a team notification.
- Handle errors so the API never crashes or returns a confusing answer.

In scope	Out of scope
• One workflow: lead-intake • POST endpoint with JSON body • Validation with clear error messages • AI classification (priority, service, summary) • Save every lead to Google Sheets • Notify the team for hot leads • Short README and demo recording	• Building the website or the form itself • Paid plans, custom domains or paid APIs • User login and authentication • Complex CRM integrations • Anything not listed under scope (ask your mentor first)

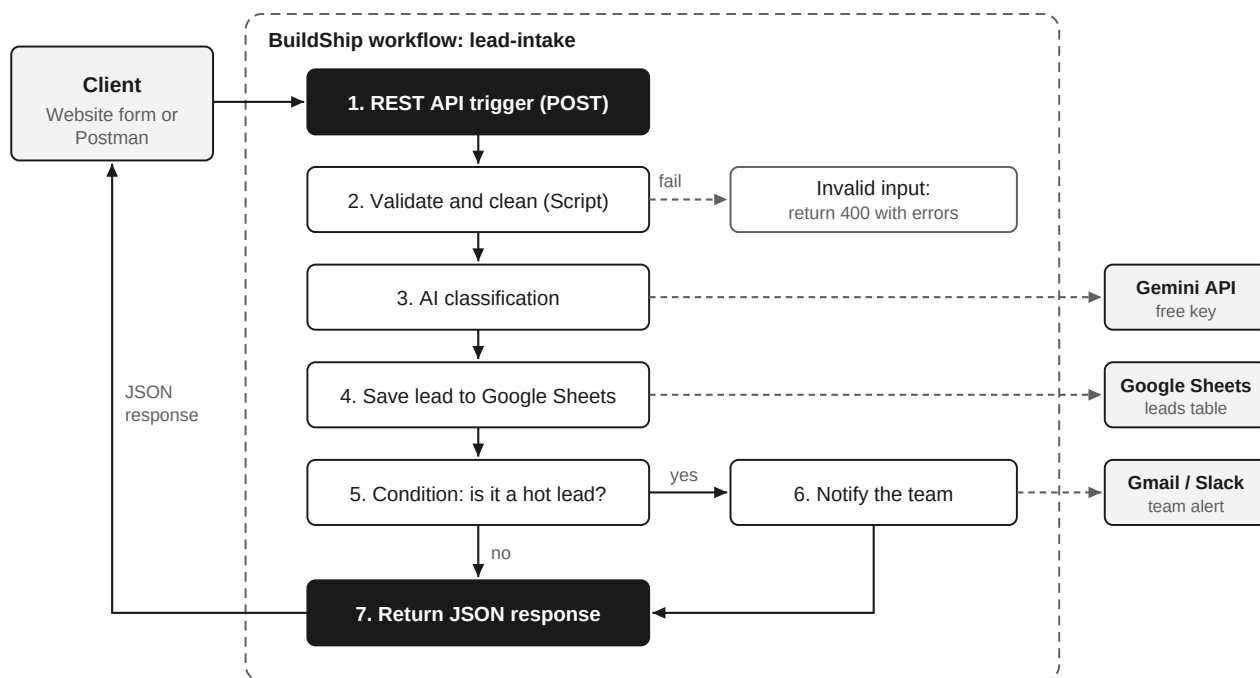
Accounts and tools you need (all free)

Item	What it is for
BuildShip free account	Hosts and runs your workflow
Google account	Google Sheets for storage, and optionally Gmail for alerts
Free-tier AI API key	For the classification step. A Google AI Studio (Gemini) key works well and has a free tier.
Slack workspace (optional)	Alternative to Gmail for hot-lead alerts. Pick one, not both.
Postman or curl	Sending test requests to your API

Check the free limits first. Free plans change over time. Before Day 1, open the BuildShip pricing page and note the current limits on workflows, runs and nodes. If anything blocks you, tell your mentor right away instead of upgrading.

System architecture

How a single lead travels through your workflow, from the website to the team inbox.



How to read the diagram

- **Solid arrows** show the main path of the request. **Dashed arrows** show calls to outside services.
- The **dashed box** is everything that lives inside BuildShip. The boxes outside it are the client and the third-party tools.
- Steps 1 and 7 are filled dark because every workflow starts with a trigger and must end by returning a response.

Why this order?

Validate before you call the AI, so you never spend AI requests on bad data. Save to the sheet before you notify, so no lead is lost even if the alert fails.

API contract and build steps

The exact request and response your workflow must support, and what each step does.

Request

```
POST /lead-intake
{
  "name": "Sara Ahmed",
  "email": "sara@acme.com",
  "company": "Acme Ltd",
  "message": "We need a chatbot
  for our store, budget ready,
  want to start this month.",
  "source": "website-contact-form"
}
```

Success response (200)

```
{
  "success": true,
  "leadId": "L-1718000000",
  "priority": "hot",
  "serviceInterest": "ai-chatbot",
  "summary": "Retail client wants
  a chatbot, budget ready."
}
```

Error response (400)

```
{
  "success": false,
  "errors": ["email is not valid", "message must be at least 10 characters"]
}
```

Build steps, node by node

#	Step	What to build
1	REST API trigger	Create a POST endpoint. Define the five input fields in the trigger schema.
2	Validate and clean	Script node. Trim text, lowercase the email, check name (2+ chars), email format and message (10+ chars). Collect all problems into an errors array.
3	AI classification	Send the message to the AI node. Ask for JSON only with priority (hot, warm, cold), serviceInterest and summary (max 20 words). Parse the result in a Script node.
4	Save to Google Sheets	Append one row per lead using the columns listed below.
5	Condition	If priority equals hot, go to step 6. Otherwise skip to step 7.
6	Notify the team	Send a short email or Slack message with name, company, summary and the lead's email.
7	Return response	Return the success JSON. If anything failed earlier, return the error JSON with the right status code.

Google Sheet columns

timestamp leadId name email company message source priority serviceInterest summary

Day-by-day plan

A suggested schedule. Day 4 is a buffer, so if you are ahead, use it for the stretch goal.

Day 1	Setup, trigger and validation • Create your accounts and explore the BuildShip dashboard and node library. • Build the REST trigger and the validation Script node. • Test valid and invalid requests in Postman.
Day 2	AI classification • Add your AI API key as a secret. Never paste keys into node code. • Write and refine the classification prompt using at least 8 sample messages. • Parse the AI answer safely and apply a fallback value if it fails.
Day 3	Storage, condition and alerts • Connect Google Sheets and save every lead. • Add the hot-lead condition and the Gmail or Slack notification. • Build the final JSON response.
Day 4	Hardening, testing and handover • Run the full test table below and fix anything that fails. • Write the README and record the demo. • Stretch goal: a second workflow that sends a daily summary of new leads.

Test cases to pass

Test	Expected result
Complete message with clear buying intent	200, priority hot, alert sent, row saved
General question with no budget or timeline	200, priority warm or cold, row saved, no alert
Missing email or invalid email format	400 with a clear error, nothing saved
Message shorter than 10 characters	400 with a clear error, nothing saved
AI call fails or returns broken JSON	200, priority "unscored", row still saved

Deliverables and evaluation

What to hand in, and how your work will be reviewed.

What to submit

- ☐ Screenshots of the full workflow canvas in BuildShip, plus the public or shared workflow link if available.
- ☐ A link to the Google Sheet with at least 8 test leads (view access for your mentor).
- ☐ A Postman collection or a list of curl commands for every test case.
- ☐ A one-page README: what it does, how to call it, fields, and setup steps.
- ☐ A 3 to 5 minute screen recording of a live demo, including one hot lead alert.

Evaluation rubric

Area	Points	What we look for
Working workflow	40	All 7 steps work end to end and the API contract matches this brief
Error handling	20	Clear 400 errors, AI fallback, no crashes on bad input
AI prompt quality	15	Consistent, structured output across varied messages
Documentation	15	README that a new teammate could follow without asking you
Demo and communication	10	Clear walkthrough, honest notes about what is unfinished

Tips for success

- Test after every node. Do not build all seven steps and then debug.
- Use the BuildShip run logs to see exactly what each node received and returned.
- Keep API keys in secrets or environment settings, never in code or screenshots.
- If you are stuck for more than 45 minutes, ask your mentor. Bring what you tried.
- Node names in BuildShip may differ slightly from this document. Pick the closest match in the library.

Check-ins. Short 10 minute check-in with your mentor at the end of Day 1 and Day 3. Final review on Day 4 or Day 5.